Practical Assignment of AIML(Mid-1)

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Set-1 Churn_Modelling_Dataset

```
  import pandas as pd
  import numpy as np
  import matplotlib.pyplot as plt
  import seaborn as sns
  from sklearn.linear_model import LinearRegression

1.df = pd.read_csv("Churn_Modelling.csv")
  print("Dataset Shape:", df.shape)
  print("Column Names:", df.columns)

2.df.sample(8)

3.avg_balance = df["Balance"].mean()
  above_avg = df[df["Balance"] > avg_balance]

  print("Average Balance:", avg_balance)
  print(above_avg)

4.geo_count = df["Geography"].value_counts()
  print(geo_count)

5.mean = df["CreditScore"].mean()
  median = df["CreditScore"].median()
  variance = df["CreditScore"].var()

  print("Mean:", mean)
  print("Median:", median)
  print("Variance:", variance)

6.sns.scatterplot(x="Age", y="Balance", hue="Gender", data=df)
  plt.title("Age vs Balance")
  plt.show()

7.X = df[["Age"]]
  y = df["Balance"]
  model = LinearRegression()
  model.fit(X, y)

8.prediction = model.predict([[40]])
  print("Predicted Balance:", prediction)
```

Set-2 Students_Dataset

```
  import pandas as pd
  import numpy as np
  import matplotlib.pyplot as plt

1.students = pd.read_csv("StudentsPerformance.csv")
  print(students.columns)

2.students.head(6)

3.print("Duplicate Rows:", students.duplicated().sum())
```

```
  students = students.drop_duplicates()

4.students[["math score","reading score","writing score"]].mean()

5.math_avg = students["math score"].mean()
  below_avg = students[students["math score"] < math_avg]
  print(below_avg)

6.students.groupby(["gender","parental level of   education"]).size()

7.students["Total_Score"] = students["math score"] + students["reading score"] +
students["writing score"]

  students.head()

8.students.sort_values("Total_Score", ascending=False).head(10)

9.plt.hist(students["math score"], bins=20)

  plt.title("Histogram of Math Scores")
  plt.xlabel("Math Score")
  plt.ylabel("Number of Students")

  plt.show()
```

Set-3 Housing_Dataset

```
  import pandas as pd
  import numpy as np
  import matplotlib.pyplot as plt
  from sklearn.linear_model import LinearRegression
  from sklearn.model_selection import train_test_split

1.housing = pd.read_csv("Housing.csv")
  housing.info()
  housing.describe()

2.X = housing[["area"]]
  y = housing["price"]

3.X = X.fillna(X.mean())
  y = y.fillna(y.mean())

4.plt.scatter(housing["area"], housing["price"])

  plt.title("Area vs Price")
  plt.xlabel("Area (sq.ft)")
  plt.ylabel("Price")

  plt.show()

5.X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

6.model = LinearRegression()
  model.fit(X_train, y_train)

7.slope = model.coef_[0]
  intercept = model.intercept_

  print("Regression Equation:")
  print("Price =", intercept, "+", slope, "* Area")
```

```
8.predictions = model.predict(X_test)
  print(predictions)

9.price_prediction = model.predict([[2500]])
  print("Predicted Price:", price_prediction)
```

Set-4 Movie_Dataset

```
  import pandas as pd
  import numpy as np
  import matplotlib.pyplot as plt


1.movies = pd.read_csv("IMDB-Movie-Data.csv")
  movies.info()

2.movies.tail(5)

3.movies_after_2015 =movies[movies["Year"]>2015]
  print(movies_after_2015)

4.avg_rating = movies["Rating"].mean()
  print("Average IMDB Rating:", avg_rating)

5.top_movies = movies.sort_values("Rating", ascending=False).head(5)
  print(top_movies[["Title","Rating"]])

6.director_count = movies["Director"].value_counts()
  print(director_count.head(1))

7.runtime_std = movies["Runtime (Minutes)"].std()
  print("Runtime Standard Deviation:", runtime_std)

8.avg_revenue = movies["Revenue (Millions)"].mean()
  high_revenue_movies = movies[movies["Revenue (Millions)"] >  avg_revenue]
  print(high_revenue_movies)

9.plt.scatter(movies["Runtime (Minutes)"], movies["Rating"])

  plt.xlabel("Runtime (Minutes)")
  plt.ylabel("Rating")
  plt.title("Runtime vs Rating")

  plt.show()

10.top_directors = movies["Director"].value_counts().head(5)
   top_directors.plot(kind="bar")

   plt.title("Top 5 Directors by Movie Count")
   plt.xlabel("Director")
   plt.ylabel("Number of Movies")

   plt.show()
```

Set-5 E-commerce_Dataset

```
  import pandas as pd
  import numpy as np
  import matplotlib.pyplot as plt
```

```
1.ecom = pd.read_csv("online_retail_II.csv")
  ecom.head()

2.ecom = ecom.dropna(subset=["Customer ID"])
  ecom.head()

3.ecom["InvoiceDate"] = pd.to_datetime(ecom["InvoiceDate"])
  print(ecom.dtypes)

4.ecom["TotalAmount"] = ecom["Quantity"] * ecom["Price"]
  ecom.head()

5.country_sales = ecom.groupby("Country")["TotalAmount"].sum()
  print(country_sales)

6.top_customers = ecom.groupby("Customer ID")
["TotalAmount"].sum().sort_values(ascending=False).head(10)

  print(top_customers)

7.negative_qty = ecom[ecom["Quantity"] < 0]
  print(negative_qty)

8.avg_amount = ecom["TotalAmount"].mean()
  print("Average Total Amount:", avg_amount)

9.above_avg = ecom[ecom["TotalAmount"] > avg_amount]
  print(above_avg)

10.plt.boxplot(ecom["TotalAmount"])

   plt.title("Box Plot of TotalAmount")
   plt.ylabel("TotalAmount")

   plt.show()
```